

WASP-80b

R-2137

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-80_200706 · created 2026-08-11T23:57:56+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459036.8069 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.232 +/- 0.067
Transit depth (Rp/Rs)^2	5.37 +/- 3.11 [%]
Orbital Inclination (inc)	86.82 +/- 1.72
Airmass coefficient 1 (a1)	38.49 +/- 16.94
Airmass coefficient 2 (a2)	-2.99 +/- 1.58
Scatter in the residuals of the lightcurve fit is	34.58 %
Best Comparison Star	#2 - [571, 417]
Optimal Aperture	1.14
Optimal Annulus	3.09
Transit Duration (day)	0.027 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.232 $\pm$ 0.067 vs 0.1714 (deviation 0.9 σ)
Duration fit vs published	0.027 d vs 0.0892 d (deviation 4.78 σ)
Mid-time O-C	-4.6 min
Depth SNR	3.5
Residual scatter	34.58 %

Light curve

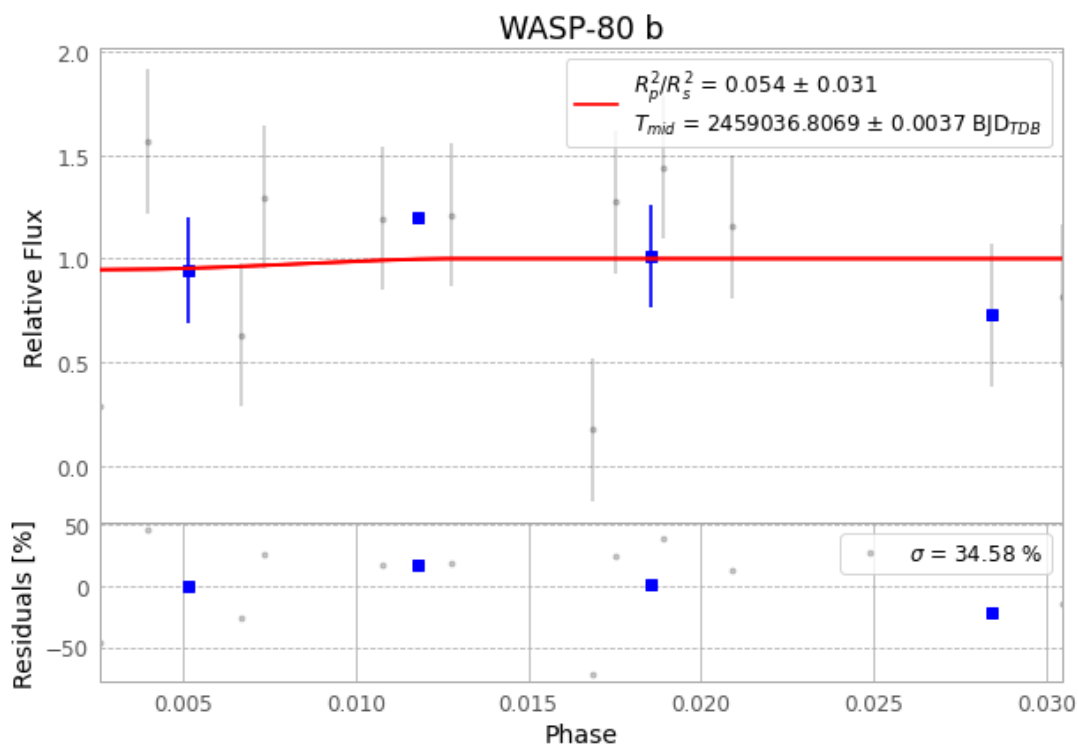


Figure 1 — FinalLightCurve_WASP-80 b_2020-07-06.png (SHA-256 8d720aeb805869bf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [417, 428], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5af55a1ad756e610...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

35 FITS frames (23 MB), WASP-80200706072413.FITS → WASP-80200706092715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-80-200706.log (ebc34d434e0b...), FinalLightCurve_WASP-80 b_2020-07-06.png (8d720aeb8058...), FinalLightCurve_WASP-80 b_2020-07-06.csv (5ffce47b85ef...)

Compute resources

Wall time 136.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 23:57Z.